

PROJECT 31A — PLATFORM DATA FLOWS BIBLE

(DOCUMENT 11)

Version 1.0.0

Classification: **CONFIDENTIAL — FOR INTERNAL DEVELOPMENT AND DUE DILIGENCE ONLY**

1. UNIFIED SYSTEM DATA LIFECYCLE

Project 31A orchestrates data flows across external APIs, local scraping caches, vector indices, relational schemas, and asynchronous event streams. Data transitions through four lifecycle phases:

- 1. **Ingestion & Validation:** Verification of domains, schemas, and credentials.
- 2. **Analysis & Vectorization:** Generating keyword volume metrics, SERP maps, and text embeddings.
- 3. **Prospecting & Outreach Routing:** Domain discovery, email deliverability validation, and personalized dispatches.
- 4. **Acquisition & Lifecycle Verification:** Reply classifications, verified link detections, and time-series snapshots.

2. CLIENT ONBOARDING FLOW & INGESTION

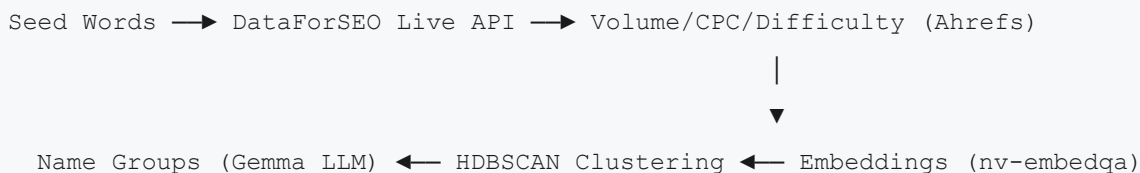
The onboarding sequence handles domain DNS checks, technology fingerprinters, and competitor discovery.



1. **DNS check:** `validate_client_domain_activity` validates DNS MX records, SSL expiration, and HTTP response codes.
2. **Business Enrichment:** `enrich_business_profile_activity` uses Scrapling and Wappalyzer to extract descriptions, contact emails, and tags, writing them to `Client.profile_data`.
3. **Competitor Discovery:** `discover_competitors_activity` calls Ahrefs to identify top competitor domains and writes the list to `Client.competitors`.
4. **Results Persistence:** `save_onboarding_results_activity` updates the database state and sets the client status to `active`.

3. KEYWORD RESEARCH & VECTORIZATION DATA PIPELINE

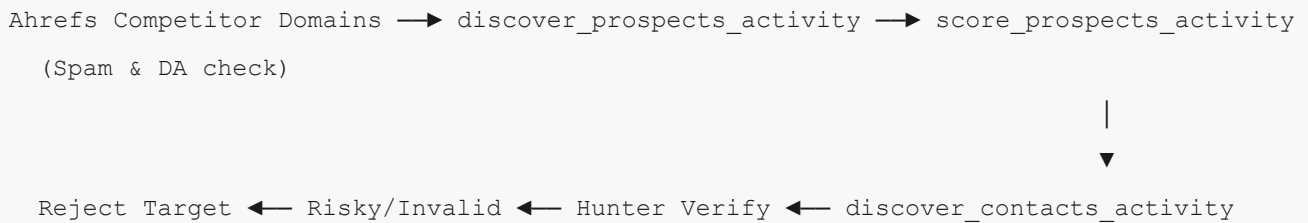
Keyword discovery runs through a multi-provider pipeline to generate metric snapshots and semantic vector collections.



1. **Expansion:** `expand_keywords_activity` submits seeds to the DataForSEO live keyword ideas endpoint.
2. **Metric Enrichment:** `enrich_keywords_activity` queries Ahrefs v3 search metrics to obtain search volumes, Cost-Per-Click, and difficulty metrics.
3. **Embedding Generation:** `generate_keyword_embeddings_activity` routes keyword strings to the `nv-embedqa-e5-v5` model.
4. **Vector Storage:** The generated 1024-dimensional vectors are stored in **Qdrant** with payload properties (`tenant_id` , `client_id`). The vector UUID is written to `Keyword.embedding_vector_id`.
5. **Semantic Clustering:** `cluster_keywords_activity` retrieves vector embeddings from Qdrant, processes them via HDBSCAN density clustering, and writes `KeywordCluster` associations to the database.

4. PROSPECTING & CONTACT ACQUISITION FLOW

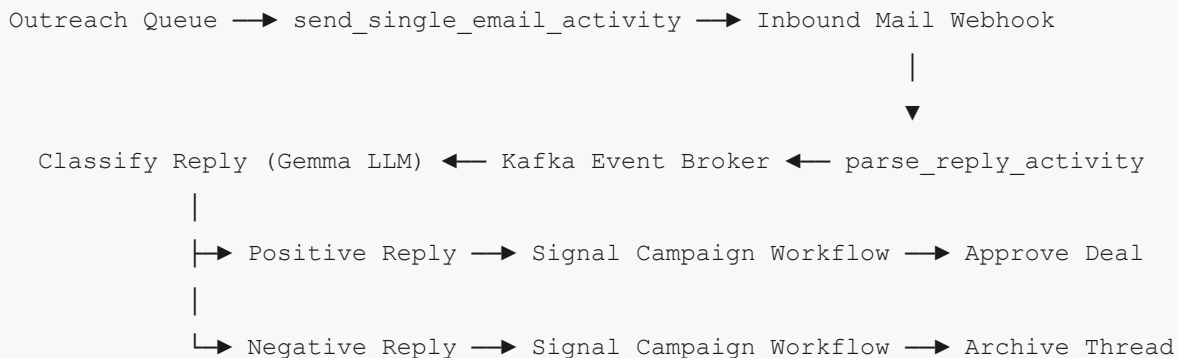
This pipeline extracts target backlink opportunities, filters out spam sites, and harvests validated contact details.



1. **Domain Intersect:** `discover_prospects_activity` calls Ahrefs referring domains endpoint for each competitor domain, outputting a list of potential target websites.
2. **Vetting & Spam Vetting:** `score_prospects_activity` fetches domain metrics and checks the domain against the link farm detector. Viable targets are written to `backlink_prospects` with status `new`.
3. **Contact Harvesting:** `discover_contacts_activity` executes a Hunter.io domain search to identify up to 3 email candidates.
4. **Email Verification:** Calls Hunter email verifier.
 - If status is `deliverable`, saves the contact email and marks the prospect as outreach-ready.
 - If `undeliverable` or `risky`, rejects the prospect.

5. OUTREACH DISPATCH & RESPONSE WEBHOOK LOOP

This loop handles email dispatch, delivery monitoring, and reply classifications.



1. **Dispatch:** The worker checks SRE kill switches, retrieves credentials from the vault, and dispatches the outreach email.
2. **Telemetry Ingestion:** The email provider emits webhooks tracking dispatches (`delivered`, `opened`). The API receives these webhooks and updates the database.
3. **Inbound Response Webhook:** When a prospect replies, the incoming SMTP server routes the mail payload to `/api/v1/webhooks/inbound-email`.
4. **Event Streaming:** The API parses the payload and publishes an event to the Kafka topic `backlink.outreach.reply_received`.
5. **Worker Processing:** The worker event consumer reads the event and launches `parse_reply_activity`.

6. Reply Classification: The activity routes the text to the `Gemma` LLM model.

- If classified as `interested`, signals the parent campaign workflow, raising an alert in the action center.
 - If classified as `not interested`, updates the thread status to `unsubscribed`.
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6. DATA RETENTION & TENANT PURGING POLICY

To maintain compliance (e.g. GDPR, CCPA) and manage infrastructure costs, the database implements a strict data retention and purging policy.

- **Data Lifetimes:**

- `audit_log`: Maintained for `365` days (1 year) using PostgreSQL partition pruning.
- `prospect_score_history`: Retained for `90` days.
- `campaign_health_snapshots`: Retained for `180` days.

- **Tenant Purging Sequence:** When a tenant is purged (`DELETE FROM tenants WHERE id = :tenant_id`), the cascading triggers execute:

1. `CASCADE` constraints drop all records in `users`, `clients`, `backlink_campaigns`, `backlink_prospects`, `outreach_threads`, and `acquired_links` automatically.
2. The purge service deletes associated vector payloads from Qdrant using the filter `{ "tenant_id": tenant_id }`.
3. The service calls MinIO/S3 APIs to delete all object storage directories matching `s3://reports/{tenant_id}/*`.
4. Records the deletion transaction in the global append-only ledger for compliance.